

Problem Statement

An organization wants to calculate employee bonuses based on different criteria.

You are required to implement **method overloading** in Java to calculate the bonus in three different ways.

Create a class **Employee** with overloaded methods named **calculateBonus**.

You need to write an overloaded method `calculateBonus` which has three forms:

1. Takes in just the salary as a double and calculates the bonus as:
 $0.10 * \text{salary}$
2. Takes in both salary and performance rating (both double) and calculates the bonus as:
 $\text{salary} * (0.05 + \text{performanceRating} * 0.02)$
3. Takes in salary (double) and years of experience (integer) and calculates the bonus as:
 $\text{salary} * (0.08 + \text{yearsOfService} * 0.01)$

All three forms should return the calculated bonus as a double.

Input Format

The first line contains a double value **salary** — the employee's basic salary.

The second line contains an integer **choice** — the type of bonus calculation to perform.

- Calculate bonus using salary only
- Calculate bonus using salary and performance rating
- Calculate bonus using salary and years of service

If **choice = 2**: The third line contains a double value `performanceRating`.

If **choice = 3**: The third line contains an integer value `yearsOfService`.

Constraints

- $0 < \text{salary} \leq 10^7$
- $0 \leq \text{performanceRating} \leq 5$
- $0 \leq \text{yearsOfService} \leq 40$
- `choice` can only be 1, 2, or 3

Output Format

The calculated bonus formatted to exactly two decimal places.

Sample Test Cases

Test Case	Sample Input	Sample Output
1	50000 2 4	6500.00
2	50000 3 5	6500.00
3	50000 1	5000.00

Hidden Test Cases (1 Mark each, Total = 5 Marks)

Test Case	Input	Expected Output
1	40000 1	4000.00
2	60000 2 3	6600.00
3	75000 3 5	9750.00
4	100000 2 5	15000.00
5	85000 3 10	15300.00
